Set up irene

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Note: *These installation instructions are not a substitute for basic knowledge of command-line instructions nor for installation guidelines of third-party software, and they come with no warranty—use them at your own risk.*

1. For

   (a) Install Git: Go here, download and follow the installation instructions.

   (b) Install Paraview: Go here, download and follow the installation instructions.

   (c) Install Docker Desktop

      i. Go here, download and follow the installation instructions

      ii. Start Docker desktop.

   (d) Download irene:

      i. Open Terminal

      ii. Type (on a network with no firewalls/restrictions)

```
mkdir C:\path\where\to\install
cd C:\path\where\to\install
git clone https://github.com/Michele-s-team/finite_elements.git
```

   (e) Crate the virtual environment

```
cd C:\path\where\to\install
docker pull michelecastellana/fenics_image
```

   (f) Start the virtual environment and enter it:

```
docker run -ti --name fe_1 `
-v "C:/path/where/to/install/finite_elements:/home/fenics/shared" `
michelecastellana/fenics_image:latest
```

(g) Make a test run:

i. Generate the mesh:

```
cd /home/fenics/shared/generate_mesh/2d/square

PARAMETERS_PATH="/home/fenics/shared/generate_mesh/2d/square"
MESH_PATH="/home/fenics/shared/generate_mesh/2d/square/solution"
rm  -rf $MESH_PATH; mkdir $MESH_PATH

python3 generate_mesh.py \
/home/fenics/shared/generate_mesh/2d/square \
/home/fenics/shared/generate_mesh/2d/square/solution
```

ii. Solve the variational problem:

```
cd /home/fenics/shared/steady_state/no_flow/

MESH_PATH="/home/fenics/shared/generate_mesh/2d/square/solution"
SOLUTION_PATH="/home/fenics/shared/steady_state/no_flow/solution"
rm -rf $SOLUTION_PATH; mkdir $SOLUTION_PATH

python3 solve.py square_a $MESH_PATH $SOLUTION_PATH
```

If everything worked, you should see something like this

```
Solving nonlinear variational problem.
  Newton iteration 0: r (abs) = 1.728e+02 (tol = 1.000e-06) r (rel) = 1.000e+00 (tol = 1.000e-06)
  Newton iteration 1: r (abs) = 7.536e+02 (tol = 1.000e-06) r (rel) = 4.362e+00 (tol = 1.000e-06)
  Newton iteration 2: r (abs) = 1.051e+02 (tol = 1.000e-06) r (rel) = 6.084e-01 (tol = 1.000e-06)
  Newton iteration 3: r (abs) = 6.041e+00 (tol = 1.000e-06) r (rel) = 3.497e-02 (tol = 1.000e-06)
  Newton iteration 4: r (abs) = 3.044e-01 (tol = 1.000e-06) r (rel) = 1.762e-03 (tol = 1.000e-06)
  Newton iteration 5: r (abs) = 1.522e-02 (tol = 1.000e-06) r (rel) = 8.813e-05 (tol = 1.000e-06)
  Newton iteration 6: r (abs) = 7.612e-04 (tol = 1.000e-06) r (rel) = 4.407e-06 (tol = 1.000e-06)
  Newton iteration 7: r (abs) = 3.806e-05 (tol = 1.000e-06) r (rel) = 2.203e-07 (tol = 1.000e-06)
  Newton solver finished in 7 iterations and 7 linear solver iterations.
Solving nonlinear variational problem.
  Newton iteration 0: r (abs) = 1.172e+01 (tol = 1.000e-10) r (rel) = 1.000e+00 (tol = 1.000e-09)
  Newton iteration 1: r (abs) = 2.323e-15 (tol = 1.000e-10) r (rel) = 1.982e-16 (tol = 1.000e-09)
  Newton solver finished in 1 iterations and 1 linear solver iterations.
Writing parameters to /home/fenics/shared/steady_state/no_flow/solution/metadata.csv...
... done.
Check of BCs:
1)
                <<(z - phi)^2>>_square = 7.81e-11
                <<(z - phi)^2>>_circle = 0.00e+00
2)
                <<(n^i \omega_i - psi )^2>>_lr = 3.15e-06
                <<(n^i \omega_i - psi )^2>>_tb = 2.40e-06
                <<(n^i \omega_i - psi )^2>>_circle = 1.01e-05
3)
                <<[mu - H(omega)]^2>>_[partial Omega] = 9.64e-01
F_circle =  [-0.37497481771499075, -0.13828199646324357, 3.833872471245247]
F_sigma_circle =  [0.0020471614495224805, -0.0010928591161240564, -0.7739040182371439]
F_kappa_circle =  [-0.37702197916451397, -0.13718913734711613, 4.607776489482391]
```

2. For

   (a) Install Git:

      i. Open Terminal

      ii. Type

```
sudo apt-get install git
```

   (b) Install Paraview:

```
sudo apt-get update
sudo apt-get install paraview
```

   (c) Install Docker Desktop: go here, download Docker Desktop and follow the installation instructions.

   (d) Download ~~irene~~ :

      i. Open Terminal

      ii.
```
mkdir /path/where/to/install/
cd /path/where/to/install/
```

iii. Run (on a network with no firewalls/restrictions):

```
sudo  git clone https://github.com/Michele-s-team/finite_elements.git
sudo chmod -R 777 finite_elements
```

(e) Create the virtual environment:

i. Open Terminal

ii.
```
cd /path/where/to/install/
sudo docker pull michelecastellana/fenics_image
sudo apt install -y qemu-user-static
sudo docker run --privileged --rm tonistiigi/binfmt --install all
```

(f) Start the virtual environment and enter it

```
sudo docker run -ti -v $(pwd)/finite_elements:/home/fenics/shared \
--name fe_1 michelecastellana/fenics_image:latest
```

(g) Make a test run with Case 1g.

3. For

(a) Install Git:

i. Open Terminal

ii. Type

```
/bin/bash -c \
"$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
/opt/homebrew/bin/brew  install git
```

(b) Install Paraview: go here, download and follow the installation instructions

(c) Install Docker Desktop

i. Go here, download and follow the installation instructions

ii. Start Docker desktop.

(d) Download ~~irene~~ :

i. Open Terminal

ii.
```
mkdir /path/where/to/install/
cd /path/where/to/install/
```

iii. Run (on a network with no firewalls/restrictions)

```
git clone https://github.com/Michele-s-team/finite_elements.git
sudo chmod -R 777 finite_elements
```

(e) Create the virtual environment:

i. Open Terminal

ii.
```
cd /path/where/to/install/
sudo docker pull michelecastellana/fenics_image
```

(f) Start the virtual environment and enter it

```
docker run -ti \
          -v $(pwd)/finite_elements:/home/fenics/shared \
          --name fe_1 michelecastellana/fenics_image
```

(g) Make a test run with Case 1g.